

Kompic Mk1 PCB — Reflow & Assembly Reference

by Ivan Porupski, 2026

PAGE 1 — QUICK REFERENCE (keep visible during reflow)

Step-by-Step Procedure

1. Verify baked paste on all pads; smear thin flux (NC-559-ASM) across pad areas — flux QFN/WSO center pads too. For the MEMS mic, flux pads **before** placing, keep flux out of the top port.
2. Place all ICs and connectors onto flux + baked paste (tweezers, magnification). **All parts go on now — single reflow pass.**
3. Photograph the populated board; **attach thermocouple** to board edge with kapton tape.
4. Place board on **cold** hot plate, run the profile below. Hands off once heating starts; let the cycle finish before handling.

Reflow Profile

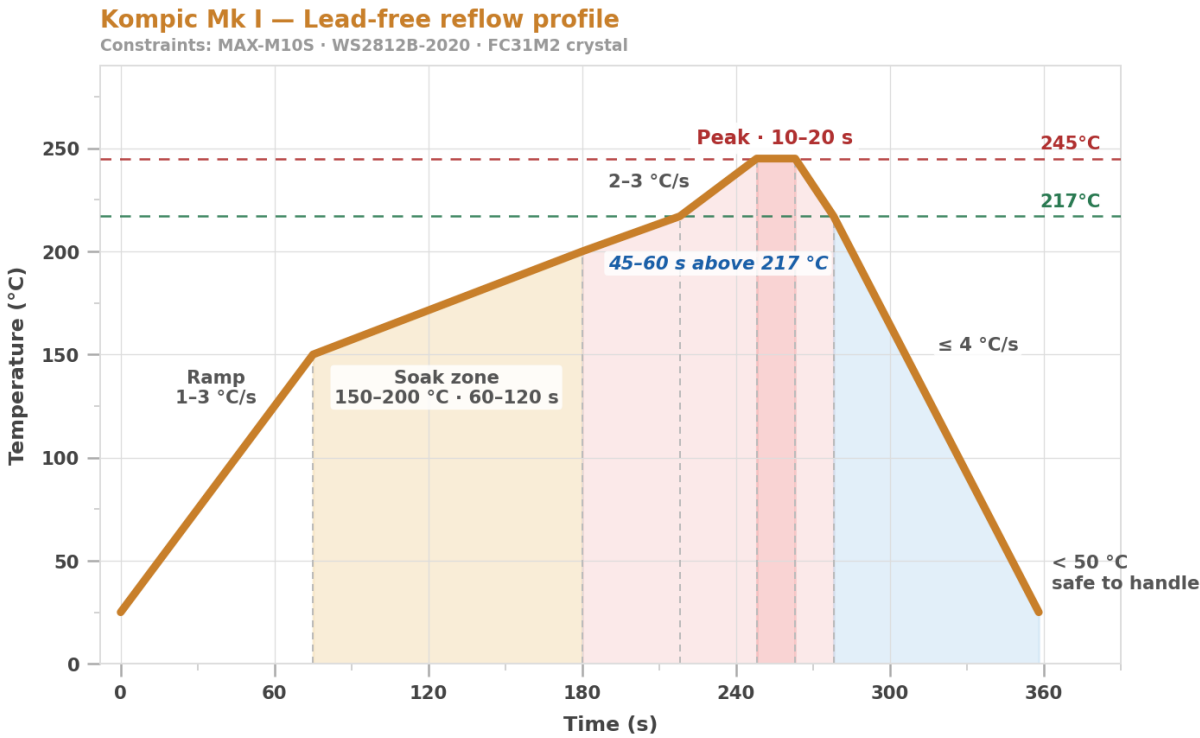


Figure 1: Target reflow profile — binding constraints from MAX-M10S, WS2812B-2020, FC31M2 crystal.

Phase	Target
Ramp to preheat	1-3°C/s, 60-90s to reach 150°C
Soak	150-200°C hold, 60-120s
Ramp to peak	2-3°C/s
Peak	245°C, hold 10-20s
Above 217°C total	45-60s
Cooling	≤ 4°C/s, hot plate off, natural cool
Safe to handle	Below 50-60°C

During Heating — Rules

- **Hands off once heating starts.** No tapping, no nudging, no adjusting.
- Below 217°C: if a part visibly shifts, you may carefully reposition with tweezers. This is a rescue, not routine.
- Above 217°C (solder liquid): **DO NOT TOUCH ANYTHING.** Surface tension self-centers parts.
- If board gets bumped above 180°C: **let the cycle finish.** Never pull board off hot plate. Fix damage after cool-down.
- Operate hot plate buttons with stable grip, controlled press, steady hand.

After Cooling

1. Visual inspection under magnification — every IC, every QFN edge, every pad
2. Compare against pre-reflow photo — check for shifted/rotated parts
3. Multimeter: resistance on every power rail to GND (3.2V, 1.8V, 5V, SYS, VBUS)
4. If all clear → initial power-on (see Page 2)
5. If short found → find and fix bridge before applying any power

Floor Life — Bag Opening (all parts, single reflow)

All MSL parts are placed in one pass, so open all moisture-sensitive bags on reflow day and reflow within their floor-life windows. Bake any part whose window was exceeded before placing.

Part	MSL	Floor Life	Bake Recovery
ESP32-S3-WROOM-1U	3	~1 week at <30°C, <60% RH	125°C / 24hr
MAX-M10S	4	72hr at <30°C, <60% RH	192hr @ 40°C, RH <5%
VEML6030	4	72hr at <30°C, <60% RH	192hr @ 40°C, RH <5%

WARNING: Single reflow pass only. The board carries bottom-side components (notably the USB-C connector) that prevent a clean second hot-plate cycle — there is no second pass for any part. Every IC, including the MAX-M10S, must be placed and reflowed together. Plan placement so nothing is missed.

PAGE 2 — BRING-UP SEQUENCE

Initial Power-On

Current-limited LiPo test:

- $4 \times 200\Omega$ 250mW resistors in parallel ($= 50\Omega$, 1W) in series with LiPo positive
- Dead short draws 74mA max — nothing smokes
- Measure voltage across board: $\sim 3.7V$ = no short, $\sim 0V$ = short present
- ESP32 will NOT boot through this resistor (needs 300mA+) — this is shorts-only testing
- Remove resistors for actual boot testing

With USB power meter arrives:

- Power via USB-C through meter
- Watch current draw in real time
- BQ25619 boots first, brings up SYS rail, TPS62840 produces 3.2V
- Normal idle current (no firmware): expect tens of mA
- Stage-2 board (charger + RTC + bus 1 sensors + mic + WS2812 + USB-C, no display / GPS / daughter): **$\sim 100\text{--}120\text{ mA}$** observed on USB-C with the stage-2 smoke sketch running (ESP32 active, sensors initialised, flashlight breathe loop). Use as ballpark, not a hard pass/fail.
- If current spikes to 1A+ immediately: unplug, find the short

NOTE: The TPS62840 is VSET-programmable. A 3.3V rail is possible; this board deliberately uses a **3.2V** rail, set by a **100k** VSET resistor (instead of 172k for 3.3V). All 3.3V-domain parts remain within spec at 3.2V.

Verification Sequence

1. Check 3.2V rail with multimeter — TPS62840 working?
2. Check 1.8V rail — XC6206 working?
3. Connect USB-C, check if ESP32 enumerates as USB CDC device
4. If ESP32 alive $\rightarrow$ I2C bus scan on bus 1 and bus 2
5. Every device should respond at expected address:

Bus 1: 0x10 (VEML6030), 0x1C (LIS3MDLTR), 0x48 (TMP117), 0x51 (PCF85063), 0x57 (MAX30101), 0x5A (CST9217), 0x6B (LSM6DSV16X), 0x76 (BME688)

Bus 2: 0x5A (DRV2605), 0x6A (BQ25619)

NOTE: The MAX-M10S GPS is **not on I2C** by default — it communicates over **UART1** (GPS-TX $\rightarrow$ GPIO17, GPS-RX $\rightarrow$ GPIO18). Verify it by opening the UART and confirming NMEA/UBX traffic, not via an I2C scan. (The M10S DDC/I2C interface at 0x42 exists as an option but is unused on this board.)

6. Missing I2C device = suspect bad solder joint on that IC
7. Open UART1 (GPIO17/18) and confirm the M10S streams NMEA/UBX — GPS solder-joint check
8. Test 5V boost (BQ25619 PMID) — enable via I2C, confirm 5V on rail

PAGE 3 — HIDDEN PAD IC VERIFICATION

These ICs have pads entirely underneath — you cannot visually inspect joints.

Part	Package	Verification Method
BQ25619	WQFN-24 (4×4mm)	Highest risk. Check I2C response @ 0x6A. Measure SYS rail voltage. Check for bridges between perimeter pads and center GND.
MAX-M10S	LCC-18	Castellated + bottom pads. Verify by opening UART1 (GPIO17/18) and confirming NMEA/UBX output. No second reflow available — get it right first pass.
BME688	LGA-8	I2C response @ 0x76. Read chip ID register.
LSM6DSV16X	LGA-14	I2C response @ 0x6B. Read WHO_AM_I register.
LIS3MDLTR	LGA-12	I2C response @ 0x1C. Read WHO_AM_I register.
MSM261DGT003	MEMS mic	No I2C — test by reading PDM data. Clock on GPIO47, data on GPIO48.
TMP117	WDFN-6-EP (2×2mm)	I2C response @ 0x48. Read temperature register (~room temp).
PCF85063ATL	DFN-10	I2C response @ 0x51. Read seconds register (should increment).

WARNING: For all hidden-pad ICs: before reflow, don't add extra paste on top of JLCPCB stencil paste. Just flux. Excess paste under QFN/LGA causes bridges.

PAGE 4 — SOLDER & CONSUMABLES

What's On The Board

JLCPCB assembles 0402 passives using SAC305 lead-free paste at $255\pm 5^{\circ}\text{C}$ (economic assembly). No passive has a peak temp limit below 260°C — no concern.

JLCPCB also applies SAC305 solder paste to ALL remaining pads via stencil (verify in their gerber viewer that paste layer apertures exist for IC pads). Paste is baked (not reflowed) before shipping. Builder reflows these pads on the hot plate at 245°C in a single pass.

Solder jumpers are pre-connected with cuttable traces — no soldering needed during assembly. Cut trace to change default state.

Your Consumables

Item	Alloy		Melting	Use On This Board
NC-559-ASM flux	—	—	—	YES — apply to baked paste, rework
Sn99.3/Cu0.7 wire 0.8mm	wire	Lead-free	227°C	YES — hand solder, rework, bridges
SAC305 paste	syringe	Sn96.5/Ag3.0/Cu0.5	217°C	YES — backup paste, rework touch-up
Sn62.8/Pb36.8/Ag0.4 paste		LEADED	181°C	DO NOT USE ON THIS BOARD

WARNING: The board arrives with SAC305 (lead-free) paste from JLCPCB. Mixing leaded paste with lead-free creates a mixed alloy with unpredictable melting point ($181\text{--}217^{\circ}\text{C}$ range). Joints form weak grain boundaries and fail as intermittent connections. **Keep all solder on this board lead-free.**

NOTE: Sn99.3/Cu0.7 wire and SAC305 paste are not identical alloys (wire has no silver, paste has 3% silver). This is fine — the small composition difference is negligible during rework. Both are lead-free, both melt in the $217\text{--}227^{\circ}\text{C}$ range.

PAGE 5 — PRACTICE RUN & TEMPERATURE CALIBRATION

Before Your First Real Reflow

You must do at least one practice run. Use a bare scrap board or one of your spare JLCPCB boards (no components).

Equipment Needed

- K-type thermocouple + reader (~€5–10) — **non-negotiable for 245°C ceiling with parts that die at 250°C**
- Kapton tape to attach thermocouple to board surface

Procedure

1. Tape thermocouple tip flat against board surface, near center
2. Run your full reflow profile using hot plate buttons
3. Log thermocouple reading at each phase vs hot plate display reading
4. Calculate offset (e.g., hot plate says 245°C but board surface is actually 252°C → set hot plate to 238°C)
5. Run again with corrected settings, verify board surface hits 245°C $\pm 3^\circ\text{C}$
6. Note your hot plate button settings for each phase — these are your calibrated numbers

What You're Calibrating

- Hot plate display accuracy (could be $\pm 10\text{--}15^\circ\text{C}$ off)
- Ramp rates (how fast does your specific plate heat and cool)
- Cooling behavior (does it cool faster or slower than 4°C/s naturally)
- Your button-pressing rhythm for each phase transition

WARNING: Without this calibration, you are guessing. A hot plate that reads 10°C low means you're actually hitting 255°C when you think you're at 245°C — and your M10S and WS2812B are over spec. With no second reflow available, a botched first pass is expensive — calibrate first.

PAGE 6 — DAUGHTER PCB REFLOW

The daughter PCB (small round board) ships attached to the main PCB but should be reflowed separately.

Components on Daughter PCB

- MAX30101EFD+T (OLGA-14, 3.3×5.6mm) — face down, HR/SpO2
- TMP117AIDRVR (WDFN-6-EP, 2×2mm) — skin temperature
- DRV2605LDGSR (VSSOP-10) — haptic driver
- Associated 0402 passives (JLCPCB assembled)

Profile

Same as main board: 245°C peak, 45–60s above 217°C, $\leq 4^\circ\text{C/s}$ cooling.

Risk Level

Lower than main board. No MSL 4 parts, no WS2812B, no crystal. All three ICs tolerate 260°C. The 245°C profile gives generous margin. This is a straightforward reflow.

Inter-Board Connection

After both PCBs are individually verified working, connect with 12 silicone wires (hand soldered to pads on each board).

PAGE 7 — REWORK PROCEDURES

Solder Bridges (most common issue)

Method: Soldering iron + flux + solder wick

- Apply flux to bridged area
- Press wick onto bridge with iron tip
- Wick absorbs excess solder
- Localized — only the affected pins experience thermal stress
- This is always the first thing to try

Misaligned / Shifted IC

If visible pads (castellated, gull-wing): reflow individual pins with iron, nudge into alignment with tweezers while last pin is still molten.

If hidden pads (QFN, LGA): must remove entirely. Use hot air rework station (controlled temp, focused nozzle). Pre-heat board on hot plate to 150°C, then hot air at 300–350°C on the IC until it lifts. Clean pads with wick + flux. Re-paste with SAC305 syringe, place new part, reflow locally.

WARNING: Exception: MAX-M10S — u-blox prohibits hot air. If M10S must be removed: pre-heat board on hot plate to 150°C, then use iron to release each castellated pad individually. Slow and tedious but possible for an LCC-18. Clean, re-paste, reflow locally.

Dead IC (confirmed bad after electrical test)

Same removal procedure as misaligned. Replace with spare from your doubles order. Localized rework only.

WARNING: There is **no second full reflow** on this board — bottom-side components (USB-C and others) prevent a clean second hot-plate cycle. All rework is **localized** (iron or focused hot air on the single failed part), never a whole-board re-reflow. Keep whole-board thermal cycles to the single production pass.

PAGE 8 — COMPLETE COMPONENT REFLOW SPECIFICATIONS

Binding Constraints Summary

Constraint	Value	Set By
Peak temperature	245°C	MAX-M10S, WS2812B-2020
Time above 217°C	45–60s	MAX-M10S $\cap$ WS2812B-2020
Time at peak ($\geq 240^\circ\text{C}$)	10–20s	FC31M2 crystal
Max cooling rate	4°C/s	MAX-M10S

Full Component Table

#	Part	Package	MSL	Peak	Key Constraints
1	MAX-M10S-00B	LCC-18	4	245°C	No hot air. Single reflow. No cleaning.
2	VEML6030	—	4	260°C	72hr floor life. Bake if exposed.
3	ESP32-S3-WROOM-1U	Module	3	235–250°C	Single reflow. Bake if open >1wk.
4	WS2812B-2020	2×2mm	—	245°C	J-STD-020C.
5	FC31M2-32.768kHz	SMD3215	—	260°C	10s at peak. Cooling $\leq 6^\circ\text{C/s}$.
6	Johanson 2450AT18A100E	Chip ant.	—	260°C	3 reflow cycles max.
7	LSM6DSV16X	LGA-14	—	235–255°C	J-STD-020 (ST TN0018).
8	LIS3MDLTR	LGA-12	—	J-STD	J-STD-020.
9	PCF85063ATL	DFN-10	—	260°C	Standard NXP.
10	BME688	LGA-8	1	260°C	50 μm min solder standoff.
11	MSM261DGT003	MEMS	—	260°C	5 reflow cycles OK.
12	MAX30101EFD+T	OLGA-14	—	260°C	Max temp only stated.
13	BQ25619RTWR	WQFN-24	—	260°C*	Assumed, TI QFN.
14	TMP117AIDRV	WDFN-6-EP	—	260°C*	Assumed, TI WDFN.
15	DRV2605LDGSR	VSSOP-10	—	260°C*	Assumed, TI VSSOP.
16	TPD2E001DRLR	SOT-553	—	260°C*	Assumed, TI SOT.
17	BSS138W	SOT-323	—	260°C*	Assumed, standard MOSFET.
18	USB-C JS16T	16-pin	—	260 $\pm 5^\circ\text{C}$	60s >217°C, 5s >250°C.

*Not stated in datasheet — standard package assumption.

MSL Handling Reference

MSL	Floor Life	Bake Recovery
1	Unlimited	None needed
3	$\sim 168\text{hr}$ (1 week) at $<30^\circ\text{C}$, $<60\%$ RH	125°C / 24hr
4	Per bag label ($\sim 72\text{hr}$) at $<30^\circ\text{C}$, $<60\%$ RH	192hr @ 40°C, RH $<5\%$

Notes

- Single reflow pass — bottom-side components (USB-C, etc.) preclude a second whole-board cycle. All parts, including MAX-M10S, go on together.
- MAX-M10S communicates over **UART1** (GPIO17 TX / GPIO18 RX), not I2C. DDC/I2C @ 0x42 is available but unused.
- All 0402 passives assembled and reflowed by JLCPCB at $255\pm5^{\circ}\text{C}$ — their process, no passives at risk.
- JLCPCB applies SAC305 solder paste to all remaining pads via stencil. Paste baked before shipping.
- Pre-soldered 0402s re-melt during builder's reflow — board must be level and flat.
- All solder on this board must be **lead-free**. Do not use leaded paste or wire.
- Paste: SAC305 (Sn96.5/Ag3.0/Cu0.5), no-clean. Rework wire: Sn99.3/Cu0.7, 0.8mm. Flux: NC-559-ASM no-clean.
- Core rail is **3.2V** via TPS62840 with a 100k VSET resistor (3.3V available with 172k; 3.2V chosen).
- Boards are **1.0mm 4-layer**.

PAGE 9 — STAGED BUILD (prototype workflow, alternative to single-pass)

NOTE: The prescribed flow is still **single reflow pass** (Pages 1–8). This page documents the *prototype* flow used on the very first board, where confidence in each subsystem was built incrementally. If you are reproducing this build from scratch and want to de-risk one ring at a time, follow these stages. Each stage has a matching smoke-test sketch under `firmware/test/`.

Stage 1 — Core (hot plate reflow)

Minimal population required to boot the ESP32 and confirm both I2C buses are healthy.

On board this stage:

- ESP32-S3-WROOM-1U-N16R8
- BQ25619RTWR charger (bus 2 @ 0x6A)
- TPS62840 3.2 V buck (100k VSET)
- XC6206 1.8 V LDO
- Flashlight LED + FET driver (GPIO41) with external pull-down
- User button (BQ_QON ↔ GPIO16)
- PCF85063ATL RTC (bus 1 @ 0x51) + FC31M2-32.768 kHz crystal + load caps
- All JLCPCB-assembled 0402 passives (top and bottom)

Excluded for this stage: USB-C connector (bottom side — keep clear so the board can still hot-plate cleanly), all bus-1 sensors, mic, WS2812, daughter PCB, display connector, GPS.

Reflow: Full hot-plate profile per Page 1 (245°C peak, 45–60 s above 217°C). Single pass on the hot plate is uncomplicated here because the bottom side is empty.

Power: No USB-C yet — apply external 5 V to VBUS pad through a current-limited bench supply, or directly to SYS for a bypass test.

WARNING: Never inject 5 V on a powered USB-C pad after stage 2 — the absent ESD diode on a missing USB-C JS16T is fine here because the connector itself isn't yet present.

Verification sketch: `firmware/test/1_smoke_test_mk1_reflow1/1_smoke_test_mk1_reflow1.ino`

Pass criteria: ESP32 enumerates as USB CDC over the programming header (or external USB-UART, since USB-C is not yet on the board); BQ25619 ACKs at 0x6A with sane STATUS/PART/FAULT; PCF85063A ACKs at 0x51 with OS bit clear (oscillator running); button toggles the flashlight; flashlight LEDC breathe visible.

Stage 2 — Sensing + USB-C (hot air, no more hot-plate cycles)

Added this stage:

- USB-C JS16T receptacle (bottom side) + TPD2E001 ESD
- VEML6030 ambient light sensor (bus 1 @ 0x10)
- LIS3MDLTR magnetometer (bus 1 @ 0x1C)
- LSM6DSV16X IMU (bus 1 @ 0x6B)
- BME688 environmental sensor (bus 1 @ 0x76)
- MSM261DGT003 PDM microphone (I2S: CLK GPIO47, DATA GPIO48)
- WS2812B-2020 signal LED (one-wire DIN @ GPIO42)

Why hot air from here on: the USB-C JS16T is bottom-side. Once it is placed, the board can no longer be put face-down on the hot plate without melting it; equally, face-up reflow risks reflowing the USB-C joints uncontrolled. **Stage 2 closes the hot-plate window for this board.** All further

additions are localised hot-air or iron rework.

Procedure:

1. Flux all stage-2 pads. For the MEMS mic, flux pads *before* placing and keep flux out of the top port.
2. Place all stage-2 ICs and the USB-C connector. Bottom-side USB-C should be done last with the board flipped on a holder, hot air at 300–350°C, focused nozzle, generous flux. Watch the mic and the QFN/LGA islands on the top side for any heat that telegraphs through.
3. Reflow each top-side IC locally with focused hot air. Don't park hot air on the WS2812 — its 245°C ceiling is a tight box even at the IC scale.

Electrical pre-checks (before powering):

- Multimeter continuity / resistance: VBUS-to-GND, 3V3-to-GND, 1V8-to-GND → no shorts.
- SDA/SCL (both buses) to GND → no shorts; bus 1 SDA ↔ SCL not shorted; bus 2 SDA ↔ SCL not shorted.
- Plug into USB-C through an inline current meter (USB-C power meter, USB tester, or bench supply set to 5 V with current limit).
- Expected steady-state draw: ~**100–120 mA** with the stage-2 smoke sketch running. Above ~300 mA at idle, or any visible smoke / smell, → unplug immediately and inspect.

Verification sketch: `firmware/test/2_smoke_test_mk1_hotair/2_smoke_test_mk1_hotair.ino`

Pass criteria: I2C scan returns the full expected stage-2 set — bus 1 = {0x10, 0x1C, 0x51, 0x6B, 0x76} and bus 2 = {0x6A}; each chip's WHO_AM_I / chip-ID matches; WS2812 colour walk is visible; PDM mic returns non-zero samples; the 1 Hz round-robin loop continues to print live values from every device without dropouts. Missing addresses point at suspect solder joints on that specific chip.

Stage 3 — Daughter PCB + storage (hot air / iron)**Added this stage:**

- Micro-SD card slot + associated routing
- Daughter PCB (small round board), reflowed *separately* on the hot plate per Page 6: MAX30101 (bus 1 @ 0x57), TMP117 (bus 1 @ 0x48), DRV2605 (bus 2 @ 0x5A)
- 12 silicone wires hand-soldered between main PCB and daughter PCB

Procedure: reflow the daughter PCB on its own (full profile, no constraints from the main board), verify it standalone with a small bench fixture, then connect to the main board via the 12 inter-board wires. Hot-air or iron the SD card slot onto the main board.

Pass criteria: I2C scan now also shows 0x48, 0x57 on bus 1 and 0x5A on bus 2; SD card mounts and writes a test file; DRV2605 buzzes when triggered.

Stage 4 — Display + touch**Added this stage:**

- CO5300 2.06" AMOLED + CST9217 capacitive touch on the 24-pin FPC
- OK-14F024-04 receptacle (J1) mated to the display ribbon

Procedure: the receptacle is hand-soldered or hot-air'd to the main board. Display panel mates to the connector — no reflow on the panel itself. CST9217 lives on the display assembly (bus 1 @ 0x5A), so it only joins the I2C scan once the panel is plugged in.

Pass criteria: CO5300 driver brings the panel up at COLMOD 0x77 (RGB888) via QSPI on the SPI2/FSPI IOMUX pins; CST9217 ACKs at 0x5A and reports touch events; panel survives a sleep cycle (DISPOFF 0x28 → SLPIN 0x10 → SLPOUT 0x11 + re-init → DISPON 0x29).

Stage 5 — GPS (MAX-M10S, deliberate last)

Added this stage:

- u-blox MAX-M10S-00B (LCC-18, MSL 4)
- Johanson 2450AT18A100E (or equivalent) chip antenna

Why last in this prototype flow: the MAX-M10S has the tightest constraints on the board (245°C peak, no hot air, no cleaning, MSL 4). On the prescribed single-pass workflow it sits in the same reflow as everything else. In this staged workflow, the hot-plate window has been closed by stage 2 — so the M10S goes on by hand, via the iron-on-castellation procedure from Page 7:

1. Bake the M10S per its floor-life recovery (192 hr @ 40°C, RH < 5%) if its MSL window has been exceeded since opening the bag.
2. Pre-heat the main board on the hot plate to 150°C (well below all the stage-2 parts' reflow points, USB-C handles this fine).
3. Pre-tin each castellated pad on the M10S footprint with a tiny SAC305 solder bump using the iron + flux.
4. Place the M10S, then reflow each castellation locally with the iron + flux + Sn99.3/Cu0.7 wire. Slow and tedious but well within tolerance for an LCC-18 with exposed castellations.
5. No solvents, no clean — M10S datasheet forbids cleaning.

WARNING: This is the *prototype* accommodation. The intended production workflow is the single-pass reflow on Pages 1–8, where MAX-M10S goes through the controlled hot-plate profile with everything else. Iron-on-castellations is a fallback and adds risk — expect to use a spare M10S if the first attempt cooks an internal joint.

Pass criteria: open UART1 (GPIO17 TX / GPIO18 RX) at 9600 baud and confirm NMEA / UBX traffic streaming from the M10S; 1PPS edge visible on GPIO46 once a fix is obtained.

Stage Summary

Stage	Process	Adds	New I2C	Smoke sketch
1	Hot plate (full profile)	Core + RTC	0x51 (b1), 0x6A (b2)	1_smoke_test_mk1_reflow1
2	Hot air (closes hot-plate window)	USB-C + bus-1 sensors + mic + WS2812	0x10, 0x1C, 0x6B, 0x76 (b1)	2_smoke_test_mk1_hotair
3	Hot air / iron + separate reflow	SD + daughter PCB	0x48, 0x57 (b1), 0x5A (b2)	— (per-chip tests)
4	Hand solder + mate	Display panel + touch	0x5A (b1)	— (display driver)
5	Iron on castella-tions	MAX-M10S + antenna	— (UART1, not I2C)	— (NMEA dump)

NOTE: If you are reproducing the prototype flow: keep a written log of what current the board draws at each stage with each smoke sketch running. Drift in that idle figure stage-over-stage is the cheapest early warning of a marginal joint.